

Unsupervised learning

Supervised Learning

- Data: labeled data, feature-response pairs
- Goal: to learn a mapping from to
- Problem type:
 - Classification
 - Regression

This taxonomy is about the **learning signal** and the **output type**. A supervised model can still be used for a confirmatory or inferential goal, such as testing a coefficient or estimating a causal effect; that is a different question about the purpose of the analysis.

Unsupervised Learning

- Data: *unlabeled* data, features
- Goal: to uncover hidden structure, patterns, clusters, ...
- Typical tasks
 - **Clustering analysis:** cluster features or observations into groups with common traits
 - **Dimension reduction:** transform data from high-dimensional space to a low-dimensional space while preserving as much information as possible
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Reinforcement Learning

- Data: a sequence of state-action pairs and a cumulative reward
- Goal: to learn a policy (a mapping from state to action) that maximizes the cumulative reward.
- Typical applications:
 - Game-playing (e.g., AlphaGo, Dota2),
 - Robotics control
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Quick Summary

Aspect	Supervised Learning	Unsupervised Learning	Reinforcement Learning
Goal	Learn a mapping from x to y	Discover hidden structure/patterns in x	Learn a policy that maximizes reward
Data	Labeled data	Unlabeled data	State-action pairs, and delayed rewards
Typical output	Classification or regression model	Clusters, low-dimensional embeddings	Policy

Clustering Analysis

- Popular techniques in analysis of complex data (e.g., genomics data)
- Aims to detecting homogeneous subgroups among the observations or the features
- Identities of the learned subgroups are unknown, but can often lead to meaningful interpretation after linked with additional information.
- We can cluster observations or features.
 - observations: clustering cancer samples to find similar cancer sub-types
 - features: clustering genes based on similar functions

Meaningful clusters

Without the groundtruth labels, how do we claim that our clusters results are meaningful?

[Link to a clustering playground](#)

Similarity and dissimilarity measures

Often we have to come up with numeric metrics to measure the similarity or dissimilarity between observations.

- Similarity measures
 - Pearson correlation
 - Spearman correlation
 - Kendall's
 - Cosine similarity ([scikit-learn pairwise metrics](#))
 - Mutual information for comparing labelings ([scikit-learn](#))
- Dissimilarity measures
 - Euclidean distance
 - Manhattan distance
 - Hamming / Jaccard distances for binary or set-like data ([SciPy distances](#))

- Wasserstein distance for distributions ([SciPy](#))
- Bottleneck or Wasserstein distances between persistence diagrams in topological data analysis ([GUDHI bottleneck](#), [GUDHI Wasserstein](#))

The key question is not "which measure is best?" but "what does it mean for two observations to be close in this problem?"

Similarity measures

When the features are numeric:

- Pearson correlation: the usual correlation coefficient; measures **linear** association between x_i and x_j .
- Spearman correlation: Pearson correlation applied to **ranked** observations.
- Kendall's τ : compares the ordering among pairs of observations.
- Cosine similarity: compares the **angle** between two feature vectors; useful when magnitude is less important than direction.

When the features are not numeric:

- Jaccard similarity: overlap between two sets, such as shared words, tags, or clicked items.
- Hamming similarity/distance: how many categorical or binary entries match.
- Edit distance: how many insertions, deletions, or substitutions separate two strings.
- Graph similarity: compare nodes by shared neighbors, paths, or graph embeddings.
- Mixed-type similarity: combine numeric, categorical, and binary features after deciding how each type should count.

Dissimilarity measure

- Euclidean distance: sum of squared difference between feature values (ℓ_2 -norm)

$$E = \sum_{l=1}^p \sqrt[p]{\frac{1}{2} \left(x_{i,l} - x_{j,l} \right)^2}$$

- Manhattan distance: sum of absolute difference between feature values (ℓ_1 -norm)

$$M = \sum_{l=1}^p \left| x_{i,l} - x_{j,l} \right|$$

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Checkout [this example](#).

Common Clustering Methods

- [Hierarchical clustering](#)
 - Sequence of solutions organized in a hierarchical tree structure, called the dendrogram
 - Maintain a hierarchical structure
 - When the number of clusters decrease, observations in the same cluster stays in the same cluster
- [k-Means clustering](#)
 - One of the most popular clustering method
 - Computationally efficient
 - No hierarchy among clusters
 - When k decreases, observations in the same clusters might be reassigned to different clusters
- And [many other methods](#)

Dimension Reduction

- Goal: to transform data from a high-dimensional space to a low-dimensional space, without losing too much information
- Versatile tools for
 - Exploratory data analysis (e.g., tSNE)
 - Data processing for downstream analysis (e.g., principal component regression)
 - Uncover true latent pattern (e.g., intrinsic dimension of neural activities)
- Typical methods:
 - [Principal Component Analysis](#) (PCA)
 - [t-distributed Stochastic Neighbor Embedding](#) (tSNE)

Study Report: Clustering

- Build a short HTML report showing how clustering choices affect the groups we find.
- Use a dataset or simulation with a clearly described clustering goal.
- Explain the data, preprocessing, similarity measure, clustering methods, diagnostics, and limitations.
- The report must be readable enough that someone can reproduce the analysis from the description and code.